

■■ SafeDev V2 — Production Manual & CLI Reference

AI-Powered Pre-Installation Security Guard for Python (PyPI) and Node.js (npm)

1. SafeDev V2 Overview & Security Boundary

SafeDev V2 is a unified, security-guarded package manager CLI that automatically scans open-source packages for supply-chain attacks, typosquatting, obfuscated payloads, and malicious code **BEFORE** allowing installation or execution. It enforces a strict **Pre-installation Security Boundary**: analyzed packages are NEVER installed, imported, or executed during security scans.

Feature	Implementation Details
PyPI Classifier	Trained XGBoost model analyzing 68 static AST and security features (Threshold: 0.38).
npm Classifier	Trained GradientBoosting model with StandardScaler analyzing 48 static features (Threshold: 0.13).
Execution Speed	Lightning-fast (~4s) via multithreaded AST parsing, wheel preference, and IPv4 resolution.
Fail-Closed Safety	Unhandled network/archive errors yield ANALYSIS_ERROR (Exit 3) — never defaults to SAFE.

2. Installation & Verification

```
cd D:\safedev\SafeDev-V2
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
pip install -e .
```

3. Python (PyPI / pip) Commands

Command / Shortcut	Description
safedev -p requests	Scan & prompt to install PyPI package <i>requests</i>
safedev -p requests==2.31.0	Scan & prompt to install specific PyPI version
safedev -p requests -y	Auto-confirm installation if security verdict is SAFE
safedev upgrade requests -p	Pre-upgrade scan & run pip install --upgrade requests
safedev uninstall requests -p	Uninstall Python package (pip uninstall -y requests)
safedev list -p	List installed Python packages (pip list)
safedev scan requests -p	Perform security scan ONLY (no install prompt)

4. Node.js (npm / Node) Commands

Command / Shortcut	Description
safedev -n express	Scan & prompt to install npm package <i>express</i>

<code>safedev install @angular/core</code>	Scan & install scoped npm package (auto-detected as npm)
<code>safedev -n express@4.18.2</code>	Scan & prompt to install specific npm version
<code>safedev -n express -y</code>	Auto-confirm installation if security verdict is SAFE
<code>safedev upgrade express -n</code>	Pre-upgrade scan & run npm update express
<code>safedev uninstall express -n</code>	Uninstall npm package (npm uninstall express)
<code>safedev list -n</code>	List installed npm packages (npm list)
<code>safedev scan express -n</code>	Perform security scan ONLY (no install prompt)

5. Exit Codes & CI/CD Status

Code	Verdict	Description & Recommended CI Action
0	SAFE	Malicious probability < threshold. Safe to install.
1	SUSPICIOUS	Elevated risk detected, or --help requested.
2	MALICIOUS	High risk threat detected. Installation blocked.
3	ANALYSIS_ERROR	Fail-Closed error (network/zip error). Installation blocked.